

ChE 381: Unit Operations Laboratory

MEMBRANE AIR SEPARATION

Rev: Fall 2024v1

Introduction

Processes based on synthetic or modified natural polymeric membranes are widespread and impact many commercial technologies. Several different types of membrane separation processes are used in chemical industries that include desalination and component gas separation. Treatment of gas streams with nonporous membranes enables the recovery of component gases without the need to use more costly approaches such as evaporation or liquefaction.

Membrane permeation is a separation process involving the selective transport of gas molecules through a permeable polymeric film. Unlike most chemical engineering separation processes which are governed by phase equilibrium relations, membrane separation is based primarily upon the relative rates of mass transfer.

Transport will occur by a solution-diffusion mechanism comprised of:

- adsorption of the gas molecule onto the interface of the high-pressure side of the membrane
- dissolution of the gas into the membrane at the interface and diffusion of the gas through the membrane
- elution of the gas from the membrane at the interface
- desorption of the gas molecule from the low pressure side of the membrane

Theory [1], [2]

Gas permeation is the term used to describe a membrane separation process using a nonporous semi-permeable membrane. In this, a gaseous feed stream is fractionated into permeate and non-permeate streams. Transport occurs by a solution-diffusion mechanism and membrane selectivity is based upon the relative permeation rates of the components through the membrane. Each gaseous component transporting through the membrane has a characteristic permeation rate that is a function of the ability to dissolve and diffuse. The two relationships upon which the equations are based are Fick's Law (diffusion) and Henry's Law (solubility). See Chapter 26 for more information.

Diffusive flux through the membrane can be expressed by Fick's Law related to the membrane system by: (*see Eq. 26.5 in [1]*)

(1)

$$J_A = D_A \left(\frac{C_{a1} - C_{a2}}{Z} \right)$$

Where:

- J_A = flux of component A in $\frac{\text{mol}}{\text{m}^2 \text{s}}$
- D_A = diffusivity of component A in $\frac{\text{m}^2}{\text{s}}$
- Z = thickness of membrane in meters
- C_{a1} = concentration of component A inside membrane wall on feed side in $\frac{\text{mol}}{\text{m}^3}$
- C_{a2} = concentration of component A outside membrane wall on permeate side in $\frac{\text{mol}}{\text{m}^3}$

The total flux of component A , J_A , can be calculated from the following expression.

$$J_A = \frac{V_{Ap}\rho}{nA} \quad (2)$$

Where:

- V_{Ap} = volumetric flowrate of species A in the permeate in $\frac{\text{mol}}{\text{m}^3}$
- ρ = density of permeate in $\frac{\text{mol}}{\text{m}^3}$
- A = area of membrane, in this case 2.7m^2 per module
- n = number of modules used

From Henry's Law: (Eqn. 26.6 in [1])

$$C_A = p_A S_A \quad (3)$$

Where:

- C_A = concentration of component A inside membrane wall in $\frac{\text{mol}}{\text{m}^3}$
- S_A = solubility constant for component A in the membrane in $\frac{\text{mol}}{\text{m}^3 \text{Pa}}$
- p_A = partial pressure of component A in the gas phase in Pa

Permeation through the membrane is a function of solubility as well as diffusivity. The product $D_A S_A$ is the flux per unit pressure gradient and is called the permeability coefficient, q_A . [1]

$$q_A = D_A S_A \quad (4)$$

Where:

- D_A = diffusivity of component A in $\frac{\text{m}^2}{\text{s}}$
- S_A = solubility constant for component i in the membrane in $\frac{\text{mol}}{\text{m}^3 \text{Pa}}$
- q_A = permeability coefficient, in $\frac{\text{mol}}{\text{m s Pa}}$. It is often expressed in Barrers, where;

$$1 \text{ Barrer} = 10^{-10} \text{cm}^3 (\text{STP}) \text{ cm/cm}^2 \text{ s cmHg}$$

See table 26.1 [1] for permeabilities of light gases in a few types of polymers.

Equations 1, 3, and 4 can be combined to give (Eqn. 26.8 [1])

$$J_A = \frac{q_A}{Z}(p_{A1} - p_{A2}) = Q_A(p_{A1} - p_{A2}) \quad (5a)$$

Where Q_A is the permeability in units $\frac{\text{mol}}{\text{s m Pa}}$, p_{A1} is the partial pressure of component A on the feedside (tube), and p_{A2} is the particle pressure of component A on the permeate side (shell). A related equation to calculate the overall mass transfer coefficient for species A is:

$$F = K_A(p_{A1} - p_{A2}) \quad (5b)$$

Where F is the species A flowrate (mole/min) and K_A is the overall mass transfer coefficient (mole/min Pa). See Example Calculation below for more details.

Separation efficiency is based on the different rates of permeation of the gas components. Data are not commonly available.

$$\alpha_{AB} = \frac{Q_A}{Q_B} \quad (6)$$

Where:

- Q_A and Q_B are defined before.

An experimental separation factor is frequently used to quantify the separation of a binary system of components can be defined in terms of mole fractions which is convenient as the oxygen sensors measure concentrations in mole percent.

$$\alpha'_{AB} = \left(\frac{\frac{\chi_{Ap}}{\chi_{Bp}}}{\frac{\chi_{Ar}}{\chi_{Br}}} \right) \quad (7)$$

And

$$\alpha''_{AB} = \left(\frac{\frac{\chi_{Ap}}{\chi_{Bp}}}{\frac{\chi_{Af}}{\chi_{Bf}}} \right) \quad (8)$$

Where:

- α'_{AB} = separation factor based on retentate composition
- α''_{AB} = separation factor based on feed composition

In these equations, the subscripts 'p', 'r' and 'f' denote the permeate (shell), retentate (tube) and feed (inlet).

Recovery is defined by the following equations:

$$O_2 \text{ Recovery} = \left(\frac{V_p C_{O_{2p}}}{V_f C_{O_{2f}}} \right) \quad (9)$$

$$N_2 \text{ Recovery} = \left(\frac{V_r C_{N_{2r}}}{V_f C_{N_{2f}}} \right) \quad (10)$$

Where:

- V_p = volumetric flowrate of permeate in $\frac{m^3}{s}$
- V_f = volumetric flowrate of feed
- $C_{O_{2f}}$ = molar concentration of oxygen in feed in $\frac{mol}{m^3}$
- $C_{O_{2p}}$ = molar concentration of oxygen in permeate in $\frac{mol}{m^3}$
- $C_{N_{2f}}$ = molar concentration of nitrogen in feed in $\frac{mol}{m^3}$
- $C_{N_{2r}}$ = molar concentration of oxygen in feed in $\frac{mol}{m^3}$

Stage cut is defined as the fractional amount of the total feed entering a membrane module that passes through a membrane as a permeate.

$$STAGECUT = \left(\frac{Q_p}{Q_p + Q_r} \right) \quad (11)$$

Where:

- Q_p = volumetric flowrate of permeate in $\frac{m^3}{s}$
- Q_f = volumetric flowrate of feed in $\frac{m^3}{s}$

The design of the experimental system is such that the concentrations and flowrates are measured at atmospheric pressure for both the permeate and non-permeate streams.

Apparatus

The experimental apparatus consists of two Prism™ separator modules (columns) that can be operated in series or parallel. Each module contains thousands of non-porous, semipermeable membranes in the form of hollow fibers. The fibers are held within a cylindrical shell, similar to a shell and tube heat exchanger.

Oxygen and nitrogen permeate the fiber walls into the cylindrical shell at different rates and are collected in a manifold at the bottom of the separator as the permeate stream. The less permeable nitrogen and residual oxygen pass through the fiber lumens and exit from the end of the separator as the non-permeate stream.

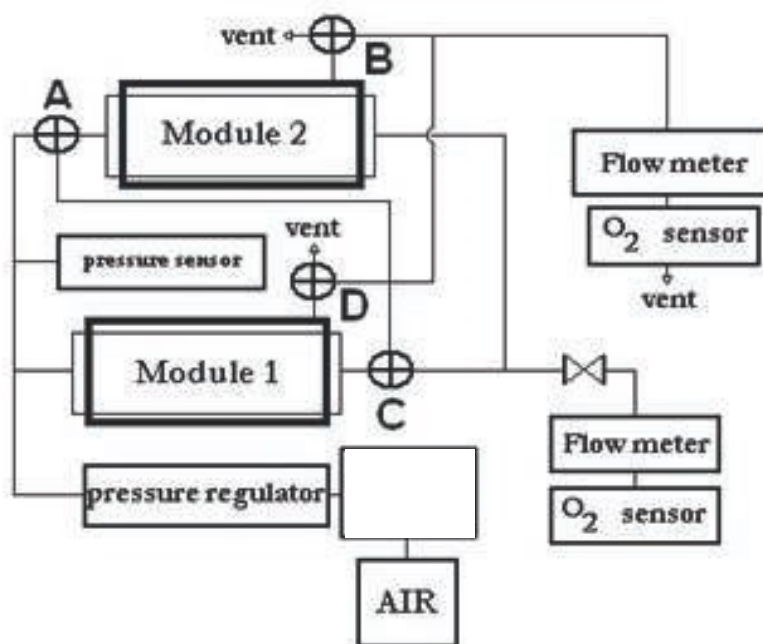
The two modules are mounted together on a common manifold. Each has approximately $2.7m^2$ of membrane area. The modules have a 1-inch diameter and are three feet in length. The polysulfone post-treated with silicone hollow fibers that comprise the

membranes have diameters of approximately 450 micrometers. Flow through them can be directed in series or parallel fashion to study the differences between configurations. The process diagram on the following page.

In **parallel** configuration the valves should be as follows:

- Valve **A**: LEFT
- Valve **B**: Up to sample module 2 shell, Down to vent
- Valve **C**: DOWN
- Valve **D**: Up to sample module 1 shell, Down to vent

Figure 1: Schematic of Membrane Separation Setup



Series configuration is the following:

- Valve **A**: RIGHT
- Valve **B**: Up to sample module 2 shell, Down to vent
- Valve **C**: UP
- Valve **D**: Up to sample module 1 shell, Down to vent

Since all piping and valves are visible, these configurations can easily be verified. **DO NOT ALLOW THE 3-WAY VALVES TO BE PLACED IN THE MIDDLE POSITION.**

Experiment:

A. SAFETY

SAFETY IS OF OVERRIDING IMPORTANCE. IT IS IMPERATIVE THAT YOU:

1. Wear eye protection at all times
2. Make sure all electrical components and metal parts are well grounded to a water pipe
3. Make sure that the air cylinder is well supported
4. The membrane modules can operate at up to 300 psig on the shell side, and this will not be reached if the pressure regulator on the cylinder is set below this pressure value. Normally, however, the shell operates at essentially atmospheric pressure. If the four 3-way valves labeled A, B, C, and D are never allowed to remain the middle position, shell pressure will be limited to one atmosphere absolute. **DO NOT ALLOW THE 3-WAY VALVES TO BE PLACED IN THE MIDDLE POSITION.**

B. Description of Experiment

The experiment is shown schematically in *Figure 1*. Two membrane modules are mounted horizontally on a 3/8 inch thick acrylic sheet. Valves allow operation in parallel or in series. Dry air from a high-pressure air cylinder passes through a 0.01 micron filter and then to a low pressure regulator. The pressure downstream from the regulator is measured by a strain-gage transducer and is displayed digitally, in the range of 0 to 200 psig. Each module contains a bundle of hollow polymeric fibers. The high pressure air feed enters the fiber lumen, corresponding to the tube side of the module. The fiber wall is more permeable to the oxygen than to nitrogen, and thus the gas leaving the other end of the fiber is depleted in oxygen while the gas from the shell side of the module is enriched in oxygen.

The flow of gas from the fiber lumens is controlled by a needle valve, and passes, at essentially atmospheric pressure, from the valve to a mass flow meter and then to an oxygen analyzer. Similarly, the gas from the shell side of the module, at essentially atmospheric pressure, passes through a flow meter and then to an oxygen analyzer. When the modules are connected in parallel, the tube side effluent streams are combined before flow and oxygen level measurement. This is also done for the shell side effluent streams. When the modules are operated in series, the valving permits the shell-side oxygen level and flowrate from each module to be measured separately.

A run is made by using the pressure regulator on the experiment to set the operating (tube-side) pressure. Then the needle valve is used to set the tube-side effluent flowrate. When a steady state has been reached, the tube-side and shell-side flowrates and oxygen levels are recorded. Thus the data for a run are:

1. Operating pressure (psig)
2. Tube side flowrate (ml/min)

3. Tube side oxygen level (%)
4. Shell side flowrate (ml/min), for module 1 and for module 2
5. Shell side oxygen level (%), for module 1 and for module 2

The ambient temperature should also be recorded, since it can have a major effect on the gas permeability of the polymeric fibers.

C. Experimental Procedure

The battery-powered oxygen analyzers (two models) should be turned on and allowed about 10 minutes to reach a steady reading. If desired they can be calibrated by gently removing the gray transducer cell from the aluminum holder and waving the transducer slowly in the air. If the percentage (by volume) oxygen indicated is not close to 20.9%, follow the calibration procedure on the O_2 sensor datasheet on the lab bench. Adjust the reading to 20.9%.

The mass flow meters are driven by a small power supply that plugs into the power strip located at the left side of the experiment. The pressure transducer display plugs into the same power strip. Note that the power strip is equipped with a ground fault circuit interrupter (GFCI), which automatically opens if a ground fault is detected. After a ground fault is repaired, the GFCI must be reset by pressing the reset button. The power strip is equipped with a 15 ampere overload protector.

Open the valve on the high-pressure air (extra dry, ultra zero compressed air) cylinder. The right-hand side pressure gauge on the two-stage regulator will show the pressure inside the cylinder. When full, it will be >2000 psig. Record the pressure inside the cylinder (Note that a typical cylinder contains about 230 cubic feet of air at standard conditions. At an average feed rate of 3 *std L/min*, this will suffice for about 40 hours of operation, and thus for hundreds of runs.)

The regulator knob can now be adjusted to step down the pressure to the desired pressure for your first experiment, typically 30-40psig. The pressure can be read on the left-hand side pressure gauge and more accurately with the pressure gauge on the apparatus. You are now ready to start your experiments.

Parallel Configuration

In the parallel configuration the valves should point as follows:

- Valve A – Left
- Valve B – Up to sample module 2 shell, Down to vent
- Valve C – Down
- Valve D – Up to sample module 1 shell, Down to vent

Since all piping and valves are visible, this configuration can easily be verified.

About 15 runs should be made, covering a range of operating pressures and tube side flowrates. Experience and available time will determine the number of runs to be made

and the corresponding conditions. Each run is ended when the flowrates and oxygen levels are essentially constant. And this will take only a few minutes. Note that a change in the tube-side flowrate, caused by a change in the needle valve setting, may affect the feed pressure slightly and require a readjustment of the pressure regulator.

A typical set of runs (series or parallel operation) might be:

Operating Tube-Side		
Run No.	Pressure (psig)	Flowrate (ml/min)
1	20	500
2	20	1000
3	20	2000
4	20	4000
5	20	8000
6	40	1000
7	40	2000
8	40	4000
9	40	8000
10	80	1000
11	80	2000
12	80	4000
13	80	8000

In general, a high flowrate on the tube side maximizes the oxygen level in the shell-side effluent, since the tube-side gas will be close to air in O_2 level. Conversely, a low tube-side flowrate will tend to minimize the oxygen content of the tube-side effluent, especially if the modules are operated with the tube sides in series. (Why?)

Series Configuration

In the series configuration the valves should point as follows:

- Valve A – Right
- Valve B – Up to sample module 2 shell, Down to vent
- Valve C – Up
- Valve D – Up to sample module 1 shell, Down to vent

A total of 15 runs should be made, covering a range of operating pressures and tube side flowrates. A typical set of runs is given above.

D. Data Analysis Example

The essential idea is to derive from the data estimates of the overall mass transport coefficients KO_2 and KN_2 through the fiber wall (membrane). We define the oxygen mass transport coefficient KO_2 by the following equation:

$$O_2 \text{ flow rate} = KO_2 (O_2 \text{ tube partial pressure} - O_2 \text{ shell partial pressure})$$

Similarly we define KN_2 on the basis of the equation:

$$N_2 \text{ flow rate} = KN_2 (N_2 \text{ tube partial pressure} - N_2 \text{ shell partial pressure})$$

The flowrates have units $gmol/min$ and the partial pressures have units $torr$. Thus the units of KO_2 and KN_2 are $gmol/(min \text{ torr})$. Note that gauge pressure ($psig$) must be converted to absolute pressure ($psia$) by adding 14.7 psi . How are these equations related to the theoretical equations (Equations 1-4)?

We can approximate the lumen (tube-side) oxygen partial pressure as the average of the inlet and outlet partial pressures. The inlet partial pressure is the operating pressure (absolute) times the oxygen mole fraction in the feed, namely 0.209. The outlet pressure is, neglecting pressure drop, the outlet mole fraction times the operating pressure. The shell side partial pressure is assumed to be the oxygen mole fraction times atmospheric pressure. The same assumptions are made for nitrogen.

The oxygen (nitrogen) transfer (permeation) rate is found from the shell side gas effluent rate ($gmol/min$) times the oxygen (nitrogen) mole fraction.

From the transfer rates and partial pressures, calculated as above, we can calculate for each run the overall transport coefficients KO_2 and KN_2 .

Sample Calculation

Where $j = 1$ denotes O_2 , $j = 2$ denotes N_2 :

P_t = tube side pressure (operating pressure) – torr

P_s = shell side pressure (atmospheric pressure) – torr

F = tube-side effluent flowrate – $gmol/min$

G = shell-side effluent flowrate – $gmol/min$

f_j = tube-side effluent flow of j^{th} species

g_j = shell-side effluent flow of j^{th} species

p_{ja} = j^{th} species partial pressure at tube inlet – torr

p_{jb} = j^{th} species partial pressure at tube outlet – torr

p_{jt} = j^{th} species average partial pressure in tube – torr

p_{js} = j^{th} species partial pressure in shell – torr

x_{1a} = O_2 species mole fraction at tube inlet

x_{1b} = O_2 species mole fraction at tube outlet

x_{1t} = O_2 species average mole fraction in tube

x_{1s} = O_2 species mole fraction in shell

K_j = j^{th} species transport coefficient – $gmol/(min \text{ torr})$

We assume that the pressure is constant with distance inside the fibers, and that the effective partial pressure p_{jt} inside the fiber is the average of the inlet and effluent pressures.

Example Calculation:

In parallel operation, assume that the tube side flowrate is 4.0 slpm (standard liters per minute), the shell side flowrate is 4.34 slpm, the operating pressure is 40 psig, the tube-side effluent is 9.8% O_2 , the shell side effluent is 29.6% O_2 .

Thus the basic data for the run are:

$$F \text{ (tube side)} = 4000/22400 = 0.179 \text{ gmol/min}$$

$$G \text{ (shell side)} = 4340/22400 = 0.194 \text{ gmol/min}$$

$$x_{1a} = 0.209$$

$$x_{1b} = 0.098$$

$$x_{1s} = 0.296$$

$$P_t = ((40 + 14.7)/14.7) 760 = 2828 \text{ torr}$$

$$P_s = 760 \text{ torr}$$

Knowing these values, we can also calculate x_{2a} , x_{2t} , x_{2s}

We then carry out, as a check on the data, an oxygen balance:

$$O_2 \text{ in} = (0.179 + 0.194) 0.209 = 0.078 \text{ gmol/min}$$

$$O_2 \text{ out} = 0.179 * 0.098 + 0.194 * 0.296 = 0.075 \text{ gmol/min}$$

The inflow and outflow are reasonably close. A similar balance can be done on the nitrogen. Next we find the partial pressures of O_2 as follows:

$$p_{1a} = 0.209 \cdot 2828 = 591 \text{ torr (tube entrance)}$$

$$p_{1b} = 0.098 \cdot 2828 = 277 \text{ torr (tube exit)}$$

$$p_{1t} = (p_{1a} + p_{1b})/2 = 434 \text{ torr (tube average)}$$

$$p_{1s} = 0.296 \cdot 760 = 225 \text{ torr (shell)}$$

Similar calculations can be performed on the partial pressures of N_2 .

The O_2 permeation rate, g_1 , is:

$$(4340 \cdot 0.296)/22400 = 0.0574 \text{ gmol/min}$$

And the partial pressure difference across the membrane is $(434 - 225) = 209 \text{ torr}$.

Thus the transport coefficient KO_2 for O_2 is:

$$\begin{aligned} KO_2 &= g_1/(p_{1t} - p_{1s}) = 0.0574/209 \\ &= 0.000274 \text{ gmol } O_2/(\text{min torr}) \end{aligned}$$

Similar calculations for N_2 give:

$$KN_2 = 0.000073 \text{ gmol } N_2/(\text{min torr})$$

The ratio of transport coefficients is 3.75, indicating the selectivity of the membrane for oxygen. Note that the coefficients above, calculated for two modules in parallel operation, are twice those expected for a single module.

E. Report

Calculate for each run the overall transport coefficients KO_2 and KN_2 , and plot these as function of operating pressure, with tube side gas flowrate and parallel/series as parameters. Discuss the trends with pressure, flowrate and configuration. What flowrate and pressure provides the best separation? Do any of the parameter have a minimal effect on the

Also, calculate and report the separation coefficients, α_{AB} , the experimental separation factors, α'_{AB} , α''_{AB} , and the O_2 and N_2 recovery (Eqns. 9 and 10) as a function of operating pressure.

References

- [1] McCabe W.L., Smith J.C., Harriot P., "Unit Operations in Chemical Engineering" 7th Edition pp. 882-922, (2005)
- [2] Koros, W.J., "Membranes: Learning a Lesson from Nature", Chem. Eng. Progr. October 1995 pp. 68-81.